Akoizo Original Practice - Form 3

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A missed reading should be entered as…

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Missing with a reason

B. Zero automatically

C. The expected mean

D. A convenient nearby value

Answer: _____

2. (2 points) The best final coherence check compares…

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. Only handwriting

B. Only the materials title

C. Question, variables, data, graph, and conclusion

D. Only page colors

Answer: _____

3. (2 points) An inconvenient point should…

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

A. Always be deleted

B. Always be replaced by the mean

C. Be hidden in the graph

D. Be reviewed transparently using evidence

Answer: _____

4. (2 points) Repeated readings on one plant always equal independent plants?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. No

B. Yes

C. Only at noon

D. Only with a calculator

Answer: _____

5. (4 points) Explain “Mean” in your own words.

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

Answer: _____
_____
_____

6. (4 points) Calculate mean and range for 8,10,12.

Answer: _____
_____
_____

7. (4 points) Give three controlled conditions.

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
_____

8. (4 points) Explain late stop timing direction.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

9. (4 points) Why show individual trials?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

10. (4 points) What should be checked in the actual packet?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
_____

11. (4 points) Write a useful follow-up question.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

12. (2 points) Define “Resolution” in the context of Experimental Design.

Answer: _____
_____
_____

13. (6 points) Solve this case and explain your reasoning: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

14. (2 points) Define “Random error” in the context of Experimental Design.

Answer: _____
_____
_____

15. (2 points) Define “Raw data” in the context of Experimental Design.

Answer: _____
_____
_____

16. (2 points) Define “Operational definition” in the context of Experimental Design.

Answer: _____
_____
_____

17. (2 points) Define “Validity” in the context of Experimental Design.

Answer: _____
_____
_____

18. (2 points) Define “Sensitivity analysis” in the context of Experimental Design.

Answer: _____
_____
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